

Organ Transplantation: making unsuitable donor organs suitable

1. General information

1.a Applicants

Name*	Institution	Department	Email	Tenured position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/ Applicant/Co-applicant]
Prof.dr. Jan N.M. IJzermans	Erasmus MC Transplant Institute	Surgery	j.ijzermans@erasmusmc.nl	Tenured	Mixed clinician/ researcher	Main lead
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Corresponding Lead: Prof.dr. JNM (Jan) IJzermans
 Planned start date of seed funding: 3-10-2022
 Primary Convergence theme: *Technology Supported Transitions in Health(care)*
 Secondary Convergence theme (if applicable): *Fundamentals of Health & Disease*
 Tertiary Convergence theme (if applicable): *Improving Health Journeys*
 Keywords: *Organ transplantation, perfusion, gene therapy, cell therapy, repair, regeneration*

1.b Summary (300/300 words)

Ageing and lifestyle changes lead to population alterations and are increasingly affecting social and health-related issues. These changes do have an impact on organ transplantation where patients of all ages need high-quality organs to be transplanted successfully. Organs retrieved from elderly, obese or unhealthy lifestyle donors and those donated after cardiac arrest do not recover from injury inherent to surgery as completely as young organs. Therefore, there is great reluctance to accept organs of marginal donors because of fear of non-function in the recipient. At present, the availability of donor organs is desperately insufficient to meet the need, and new strategies are needed to solve the donor organ shortage.

For some, living organ donation may be a successful alternative but finding a relative to donate might not always be an option as for lung and heart donation. Most patients will remain dependent on deceased organ donation, facing a 20% mortality rate on the transplant waiting list. Alternative options like dialysis, and pulmonary or cardiac support may be offered albeit with increased risk on morbidity and early death. Obviously, the current situation necessitates the development of new strategies.

In this Flagship, we envision technological innovation to build a long-term machine perfusion platform to not only sustain but also recover donor organ viability out-of-the-body. Damaged donor organs will be made suitable for transplantation. The Erasmus MC is needed to deliver fundamental insights in cellular and tissue repair, targets for pharmacotherapy and stem cell knowledge for regenerative medicine approaches to preserve, repair and regenerate organs for transplantation. A breakthrough technological innovation from TU Delft is needed to integrate machine components, optimize regulatory feedback systems, and miniaturize the system, while EUR expertise is needed to compare the outcome of these developments with alternatives in terms of costs, (health-related) benefits, patient/societal preferences, and ethical aspects.

1.c Public Summary (max 300 words)

Voor falende organen, zoals hart, longen, nieren en levers, is orgaantransplantatie de enige lange termijn oplossing. Het donortekort zal met de veroudering en levensstijl van onze bevolking toenemen (in 2040 is 25% ouder dan 65 jaar). Dit heeft grote gevolgen aangezien (1) de vraag naar orgaantransplantatie groter zal worden door toename van levensverwachting, terwijl de beschikbaarheid van donororganen op dit moment al ontoereikend is om aan de huidige behoefte te voldoen; (2) de verouderingsevolutie niet alleen zal leiden tot toename van patiënten met orgaanfalen, maar ook van kwetsbare donoren met minder geschikte organen voor donatie en (3) veranderingen in levensstijl met overgewicht, roken en alcoholgebruik ook de kwaliteit van organen ook op jongere leeftijd zal verminderen en de donorpool zal beperken. Deze huidige situatie vereist de ontwikkeling van nieuwe strategieën om het tekort aan donororganen op te lossen.

Een radicaal nieuwe oplossing voor dit probleem is om de functie van donororganen buiten het lichaam te testen met behulp van orgaanperfusiemachines en om aanhoudende schade vóór implantatie in de ontvanger te herstellen. Dus ongeschikte donororganen geschikt maken voor transplantatie. Dit is precies wat dit Vlaggenschip Orgaantransplantatie wil realiseren. Hoe? Door unieke expertise te combineren en een integrale aanpak van het Erasmus MC (biomedische kennis), TU Delft (technologisch kennis) en de Erasmus Universiteit (vaststellen economische, ethische, legale, en maatschappelijke impact van nieuwe oplossingen) te gebruiken.

Dit vlaggenschip biedt niet alleen mogelijkheden om meer patiënten op de transplantatie wachtlijst te helpen, maar het zal ook de state-of-the-art naar een punt brengen waarop een drastische toename van het aantal orgaantransplantaties haalbaar is.

2. Relevance to Convergence Health & Technology mission

(50%) — max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s) (617 words)

BACKGROUND

In this Organ Transplantation Flagship we will evaluate a fundamentally new strategy using machine-assisted organ repair to improve the quality of donor organs and reduce the chronic shortage facing the transplant community. This plan primarily contributes to the convergence theme "Technology-supported transition in healthcare" through a versatile breakthrough technology that will: 1) allow better preservation of *all donor organs*, 2) enable pharmacotherapeutic and ATMP (Advanced Therapy Medicinal Products) based repair of organs that are now considered too damaged to transplant, and 3) support regenerative stem cell approaches to even grow organs anew in the long term. The latter two goals further contribute to the theme "Fundamentals of health and disease".

CONTRIBUTION TO THE H&T MISSION:

We will address a chronic shortage in viable donor organ whilst also improving the survival of organs post-transplant. This will offer significant contribution to **lifelong health** by increasing patient accessibility to care. Every organ transplantation program has age limits for recipients to be admitted to the waiting list, while transplantation markedly increases survival in all age groups. Also older patients with end-stage organ diseases should be able to benefit from transplantation, largely restoring their expected life span and improving their quality of life.

Global **socioeconomic inequalities** in organ transplantation permeate throughout all aspects of the process. Individuals with lower income, education, or occupational status experience worse health, increasing the risk of organ failure (1). Socioeconomic status-related inequalities exist regarding both access to the waiting lists, and transplantations on listing conditionally (2). Beyond addressing these pain points by increasing availability of donor organs, the flagship will focus on repairing and regenerating organs to supply minority groups which are underserved by matching organ donors (3).

Per definition, organ transplants are the epitome of **personalized medical treatment**. With our breakthrough platform, we will make organ transplants more accessible to more people. The consortium aims to achieve organ proteome alteration to convert non-compatible organs to match with the recipient patient. These highly personalized medical approaches - which build on recent breakthroughs in the field - will be validated within our work plan to lay a foundation for their tailored implementation in clinical practice.

OBJECTIVES FIT WITH H&T THEME:

The objectives of the program most closely fit with the theme of **technology supported transitions in healthcare**, since a key objective is to develop a platform perfusion machine for organ preservation, repair and regeneration. To achieve this objective, we will involve an interdisciplinary community of technical, medical and societal researchers. In-line with the theme guidance, we will address transitions in 1) *personalized medicine* using ATMPs, 2) *integrated care* by developing a

platform system for all organ types. Towards the overall aim, this flagship will contribute to a more resilient health system by reducing the current organ shortage. To guide creation of a *value-based* and *affordable* healthcare system, the flagship employs health economic, ethical, legal and societal assessment of the novel strategy. Ultimately, we will unlock a radically new model for treatment in the field organ transplant to promote and maintain health for all. This approach will reduce waiting list times and enable patient to receive the treatment that they desperately need, **improving patient journeys** by facilitating faster resumption of daily life activities and a longer time span of healthy and active living.

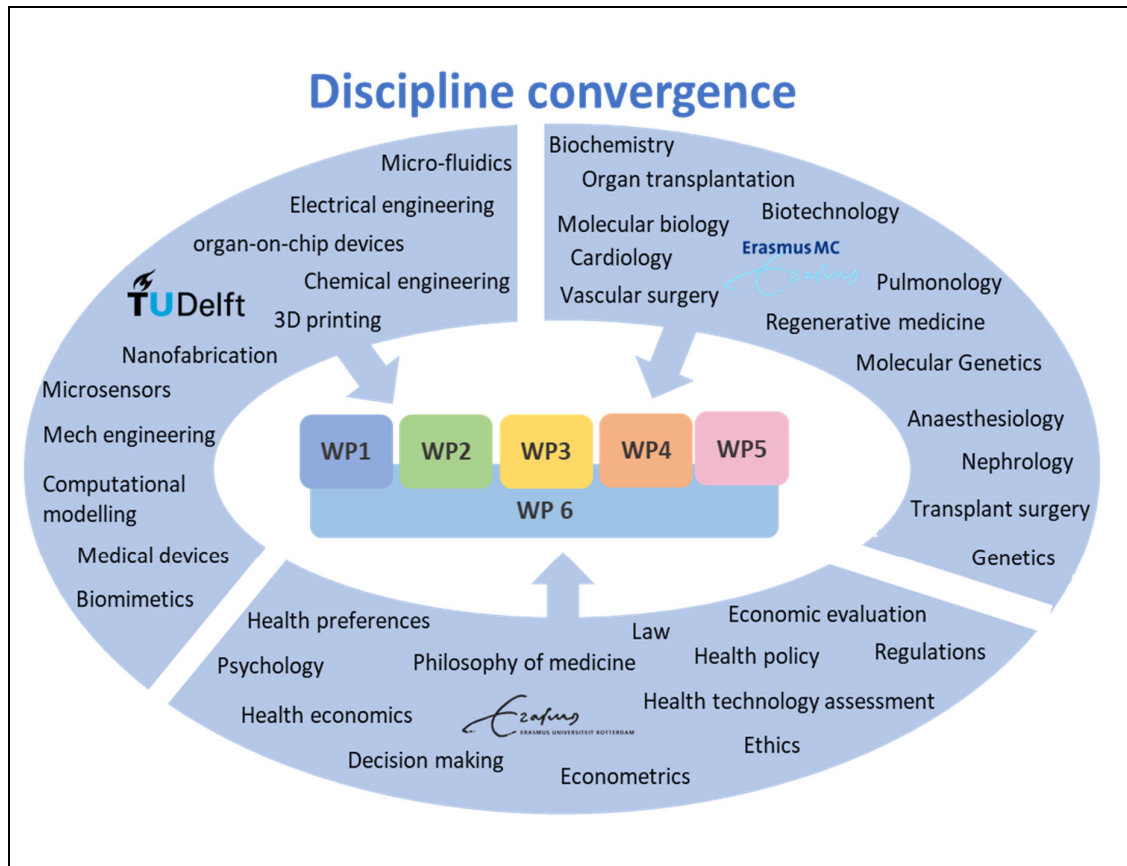
This Flagship program's objectives also fit with the guidance for the theme **Fundamentals of Health & Disease**. A core research focus will be on organ *repair and regeneration*, which will lead to the elucidation of *basic knowledge on the molecular mechanisms of health and disease*. To achieve this, novel analysis methods and instrumentation will be developed which will ultimately enable the creation of revolutionary new pharmacotherapy and ATMP based treatments to repair and regenerate organs.

2.b Synergy between disciplines (318)

The three Knowledge Institutes TU Delft, Erasmus MC and EUR will collaborate to combine their specialities for successful exploration and implementation (Figure 1). At present a long-term platform solution to treat organs outside the body does not exist (de Jonge/Ijzermans/Duncker). Although vital components are available, a breakthrough technological innovation from TU Delft (French/Ottens/Zadpoor) is needed to integrate these components, optimize regulatory feedback systems, and miniaturize the system. Furthermore, new techniques are required to monitor injury and repair processes inside the organs in the perfusion system (Essers/Ottens/Boukari/Mastrangeli). Thus, the consortium brings deep expertise in biomechanical engineering, microelectronics, precision microengineering, electronic components, technology, chemical engineering, and materials. Critically, organ transplant professionals from Erasmus MC Transplantation Institute will support this research and development, to guide TU Delft on the requirements of organs within a perfusion system. In addition, the Erasmus MC will deliver its fundamental insights in cellular and tissue repair, targets for pharmacotherapy and stem cell knowledge for regenerative medicine approaches to preserve, repair and regenerate organs for transplantation. The knowledge gained in this project will reflect on other disease applications, where tissue repair or regeneration is also needed.

Through all aspects of the project, the technologically innovative strategy will raise important societal questions about prioritization, health-related quality of life, well-being, cost-effectiveness, acceptance rate, and moral issues. As such ethicists, legal scholars, psychologists, and health economists from the Erasmus MC and the EUR and the interfaculty Erasmus Choice Modelling Centre will be involved at each stage to perform the relevant and necessary assessments (deBekker-Grob /Buijs/Schermer/IJzermans). These assessments will be critical to both inform and guide the project to ensure clear added value whilst convincing the wider community on the benefits of this platform; widespread acceptability will be key to successful market implementation and realization of the intended healthcare and socioeconomic impacts. *Figure 1* illustrates the broad and extensive range of disciplines.

Figure 1: Discipline synergy and convergence within this Flagship



2.c Long term sustainability plan (479)

The structure of the flagship project has been designed with feasibility and long-term sustainability at the focus:

Project design & activities: the work plan has been split into different themes, all with different levels of technology readiness: preservation, repair, and regeneration. In doing this, the plan is naturally sustainable given there have been set clear project milestones ranging from organ preservation in the near term (2-years), right through to organ regeneration on the 15-year horizon.

Novel collaboration: Working closely together in this Flagship, strong interdisciplinary connections will be forged and serve as catalyst for future research on organ transplantation. This convergence research will foster collaborations to increase the long-term sustainability and feasibility of the Flagship. Patient perspectives are key to the implementation of novel healthcare strategies, and thus the consortium will develop a patient council that will be involved in each step of the Flagship.

Education and partner network expansion (section 2d) will also play an important role to ensure self-sustainability and growth. Educational programs are a 'grassroots' approach to develop and inspire the next generation of scientists' understanding and involvement in the field. Industry partnering will promote continued platform development and financing, whilst collaboration with patient organizations, policy makers, and health insurers will ensure successful market uptake.

Financing: As shown in the budget and section 4a, the PIs have an extremely strong record in obtaining funding. The Flagship aims to deliver a commercially viable platform solution with strong market potential. There is also strong exploitation potential for individual technologies developed within this project (e.g. cell and gene therapy approaches). This commercial opportunity will be leveraged to attract more investment from non-dilutive programs but also private investment e.g. angel investors, VC investors, industry partnerships.

Table 1: Key risks facing the long-term sustainability of this flagship

Risks	Mitigation (words 346)
Challenges meeting regulatory standards	Regulatory experts will be engaged early to define the best regulatory strategy to avoid delays in development and commercialization. We will also engage with relevant stakeholders such as the EMA, to empower translation and patient access of our innovations.
Delayed entry of the perfusion platform approach to clinical guidelines.	Safety and efficacy data in line with regulatory requirements. These results will be communicated and disseminated to a broader audience including policy makers to aid their evidence-based recommendations for entry of the intervention to clinical guidelines.
Low societal acceptance	WP6 focuses on addressing ethical, health economic, legal, societal issues surrounding the novel organ transplant strategy in order to increase the acceptability of the platform. To maximize uptake, a sound stakeholder engagement plan will be developed including marketing activities.
Lack of follow-up funding after the CHT flagship	The consortium will seek further non-dilutive funding (NWO grants/ Horizon Europe/ EIC) as well as private investments. Industry partners (pharma, biotech, medtech) will also be engaged raise additional funding for development, with view for potential future technology licensing deals or acquisition.

2.d Visibility, attracting talents and new partners (378)

To increase the flagship visibility, the consortium will follow the communication measures described in Table 2, which revolve around 1) organ transplant challenges, 2) project progress and 3) new knowledge generation.

Table 2: communication and dissemination plan

Measures to increase visibility, attract talent and new partners	
Target group and objective(s)	Tools and/or channels
Students Stimulate early interest in the field and generate/strengthen an inflow of talent through internships and PhD programs.	Educational programs provided by Erasmus MC, TU Delft, LUMC and ESHPM – Focused courses within Nanobiology (BSc, MSc), Clinical Technology (BSc), Technical Medicine (MSc), Health Economics, Policy & Law (MSc). Information market program – Minor and master's students can virtually meet with research groups to stimulate their interest and choose a group with research objectives that aligns with their interest.
Research groups Attract additional collaboration from experts in the field from universities/ institutes throughout Europe.	Top level publications in open access journals Symposia/conferences

End-users (Clinicians and Patient advocacy groups) Obtain feedback to guide our research (as part of ethics-guided research), as well as increase engagement.	Stakeholder meetings, focus groups, social media, website, videos
Scientific organisations Broaden network of influence and attract talent and partners for further collaborations.	Presentations at scientific meetings including national and international forums (The Erasmus MC Transplant Centre belongs to the top 5 centres in Europe)
Industry Explore opportunities for economic valorization of project results, and to attract funding.	Social media (e.g. LinkedIn), Business events, Symposia/conferences.
General public Create visibility for the Organ transplantation flagship, and public acceptance for the novel transplant strategy	Social media, website, videos, Popular science articles in newspapers, television (talk) shows, TEDx talks, science festivals, and science cafes.
Government and regulatory bodies/Policy makers Facilitate translation of project results through approvals and reimbursements, and to attract future funding through grant and subsidy applications.	Symposia, specialised meetings, science festivals.

Excellent early-career researchers will play a fundamental role in this project by supporting the PIs in executing on the WP activities. The requested convergence budget includes hiring of 6,1 FTEs, 3,7% of which are PhD students and 1.0 Fte Post-Doc and in addition a quarter master will be appointed. We will also identify and collaborate with excellent early-career researchers whose fundamental work could significantly expand the medical and industrial potential of the organ transplantation platform.

2.e Long term societal impact (671)

Table 4: An overview of the long term scientific, economic, and societal impacts

Scientific impact
- The creation of a universal perfusion platform for long term out-of-the-body organ perfusion for all transplantable organs. - Fundamental new insights in cellular repair processes, that benefit other disease conditions. - Applied technological advance in (micro) sensor systems to detect organ function and viability. - Unique knowledge how ATMP (stem cell) therapies contribute to organ and tissue repair.
Economic impact
- A cost-effective alternative for (end-stage) chronic heart, lung, liver, kidney diseases and diabetes. Although transplantation itself is an investment in the first year, it has the potential to avoid high costs of chronic organ disease for many years.

- Transplantation can fully restore ability to earn income, ability to carry out caring tasks and improves quality of life significantly, both in physical and mental quality of life.
- Development of the technology will allow Dutch SME and spin-off companies to acquire pre-competitive knowledge with considerable leverage on their R&D. Application will both create direct economic value and propel further research on organ repair.

Societal impact

- Improved treatment access by shortening of transplantation waitlists for all organs.
- Zero waitlist mortality.
- Increased access for elderly and disadvantages patients (minority groups) to receive a donor organ.
- Home monitoring for patient self-management
- Revision of regulatory frameworks regarding organ transplantation.

Impact pathway

In a 5-year time frame, we expect to deliver a working long-term perfusion platform that can host all transplantable organs and sustain viability for at least 72 hours within the clinical environment of the Erasmus MC (TRL 7). In the following five years, a demonstration model will be further produced, and commercial partners explored for manufacturing.

With respect to regeneration, more fundamental research is needed and TRL 7 is aimed for on the 10-year horizon. For successful clinical translation, it is crucial that fundamental researchers, technicians, clinicians, health economists and societal researchers work together to merge their specific detailed knowledge to achieve the highest levels of readiness.

To maximize the impact on science, technology and society, the development and appropriate protection of IP will be a central component. All data generated will be published in open access journals and data will be shared via the EOSC.

Societal and economical valorisation strategies

To bring the organ perfusion platform successfully to the clinic and foster valorization, all components will be validated in the laboratory and an industrial partner will be engaged to initiate clinical grade assembly of the different components of the perfusion system, necessitating CE mark approval and adherence to the Medical Device Regulation of the EU.

Health economists, psychologists, ethicists, and health lawyers need to discuss with patient advocates and policy makers how to balance benefits and costs, ensure safety, justice and bodily autonomy and the legal interpretation of complex regulations of ATMP, if applied outside the body to an organ. Also, the consortium will engage health insurers and governmental healthcare departments (see letter of intent) from the start of the flagship to increase awareness and pave a route for the organ transplantations entry to clinical guidelines and insurance reimbursement.

In its capacity as the largest transplantation center in the Netherlands, Erasmus MC will be an end-user of this technology, further supporting the pathway to impact. As seen in the annexed letter of intent, the institute expresses strong support for this program and intends to implement the platform in clinical practice. The EMC Transplant Institute envisions development of a repair and regeneration organ hub, where other transplant centres can send their transplant organs for optimization, after which the organs can be resent to provide care close to home. This centralization of infrastructure and knowledge will facilitate efficient use of scarce resources and human capital. Furthermore, the Flagship will advocate that the required technological infrastructure, such as a GCP-rated ATMP production facility, will be implemented in the skills facility of the new Med Tech campus at Erasmus MC, enabling new medical entrepreneurship, highly likely reducing overall costs and strengthening synergy between partners.

1. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions (180 words)

Ambition

This flagship addresses **chronic donor organ shortage**, leading to waiting lists mortality rates of ~20%. The shortage is increasing due to both rising demands for replacement organs and decreased donor supply, driven by reduced donor organ quality due to life-style factors, including aging, obesity, and alcohol consumption. A radically innovative potential strategy is needed to solve this shortage.

To realize this mission, we will work towards the following goals:

- 1) Better preservation of donor organs in an *ex vivo* perfusion system replicating physiological conditions.
- 2) Pharmacotherapeutic and ATMP based repair, for organs considered too damaged to transplant.
- 3) Regenerative stem cell approaches to grow organs anew in the long term.

Innovative & challenging aspects

Our vision is to create a backbone that will support organ therapy for all transplantable organs, to develop innovative technology universally employable to detect organ viability, monitor repair and to incorporate novel cellular repair techniques to facilitate generic regeneration processes, without losing sight of its economic, ethical, legal, and societal implications. This presents numerous challenges throughout that will only be overcome through effective discipline convergence.

3.b Aims and feasibility (465 words)

Program aims

[WP1] Development of a breakthrough organoid-on-chip system allowing the *in vitro* study of organ damage and repair

[WP2] Delivery of cutting-edge perfusion platforms for out-of-the-body organ therapy.

[WP3] Pharmacotherapeutic interventions to preserve and repair donor organs using perfusion on pre-transplant and post-transplant organs

[WP4] Successful ATMP applications to repair and regenerate donor organs, to improve viability pre-transplant and to extend organ survival post-transplant.

[WP5] Development of non-invasive new technology and novel technological applications to determine organ function and viability (organometrics).

[WP6] Empirical evidence and recommendations on health economic, ethical, legal and societal implications of this advanced high-tech modular body part replacing health care.

Research plan

Budget: Convergence funding will be used to appoint researchers (master students, PhD's & post doc) to realize the aims of the WP's. Provisional posts are included for material costs and skills facilities. For details, please see budget [template](#).

Personnel: To realize the research plan 1 post-doc will be assigned to WP1-2 (TU Delft), 1 PhD to WP3 (EMC) and 2 PhD's to WP6 (EUR). A Quarter Master (0.6fte) will be appointed to coordinate

new research, education and funding initiatives. Ongoing collaboration in education between TUD and EMC (Master Medical Technology, Bachelor and Master Nanobiology), EMC and EUR ESHPM, and Convergence framework contacts will be used to select talented candidates.

Technical feasibility:

Recently, short-term cold and warm machine perfusion was introduced to extend organ quality. We will further develop these **technologies (4,5)** to extend perfusion time from 6-8 to 48 hours and, subsequently 8 days, enabling organ testing, therapeutic interventions and regenerative responses. Recent gene editing and cell-based breakthroughs support the technical feasibility for ATMP based repair and regeneration of organs, e.g. *ex vivo* perfusion mediated gene therapy to turn the organ immunologically invisible and improve post-transplant graft survival (6).

Technical risks:

Table 5: risks and contingency plan

Risk and contingency plan				
#	Risk	WP	H,M,L	Contingency
<i>Technological</i>				
1	Defining cell differentiation cues for multi organ-on-a -chip models	1	L	Elaborate on proof-of-principle studies
2	Integration of sensor technology enabling long-term perfusion	2	M	Use clinical grade perfusion system to validate sensor outcome
3	Medical ethical reticence to use manipulated donor organs	2	L	Acceptance by society will be explored in early stage
4	Balancing rejection vs drug-induced toxicity for repaired donor organs	3	L	post-transplant drug level monitoring and AI adjusted dosages
5	Use of cell-based therapies and side effects	4	M	Starting by using autologous cells

External funding plan: Existing consortium member research funding will provide €800k matching (see annex). We will apply for funding by the NSN, Heart Foundation, MLDS, Zon MW grants, VWS and European Horizon; in addition, an important factor for generating resources is valorisation, including IP protection. Devices and diagnostics at the TRL6 level will be presented to commercial parties, or start-ups may be generated.

3.c Work plan (1281 words)

Figure 2: GANTT chart

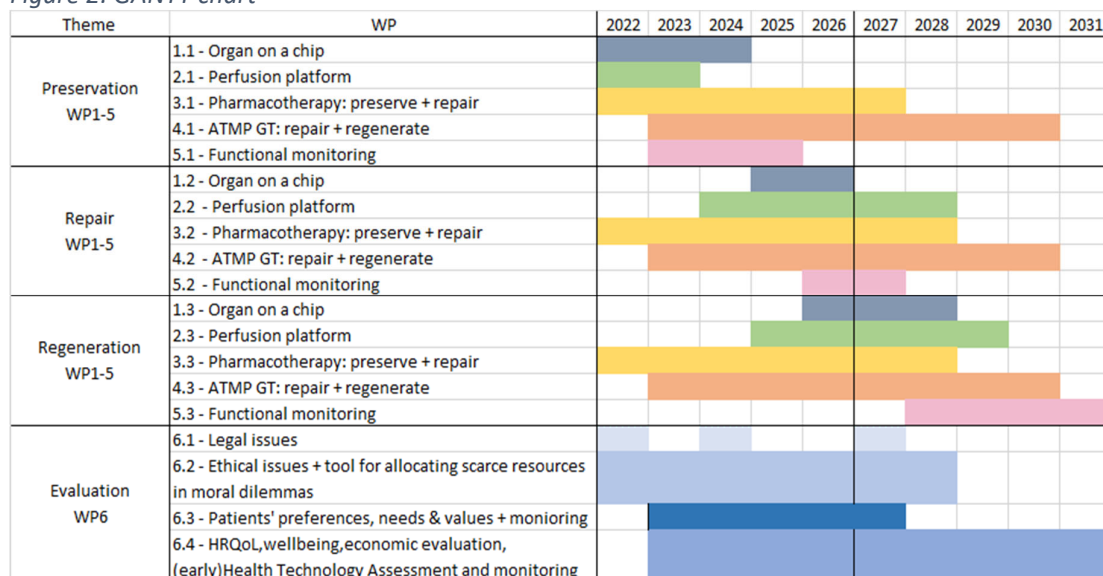


Figure 3: Pert chart; WP interaction towards the flagships long-term goal

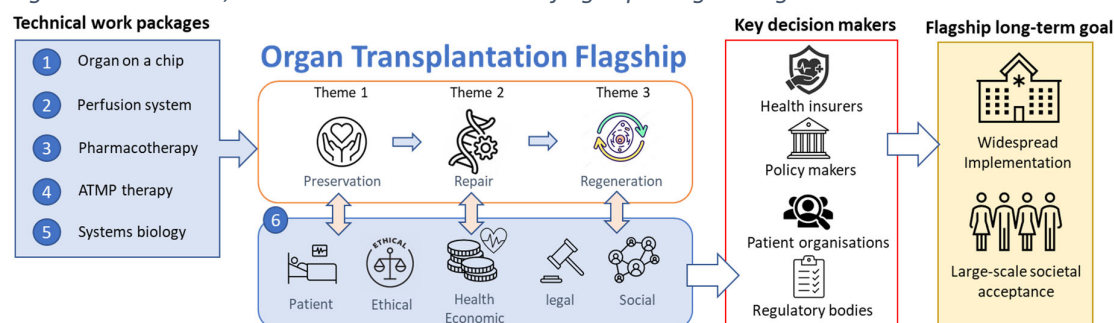


Table 6a -6f: Work packages 1 to 6

WP1	Development of a multiorgan-on-chip model for studying organ injury and repair	Total Duration	2022-2029
Institute(s)	TU Delft/Erasmus MC		
Summary	Organoid graft-on-a-chip models will lead to new insights for the development of perfusion platforms (WP2), advance the development of repair and regenerative therapies for transplant organs, and enable studying of immunological interactions, and responsiveness to immunosuppressive and regenerative therapies (WP3 & WP4).		
Theme 1	Preservation	Duration	2022-2025
Objective/Activities	Development of multiorgan-on-a-chip model to study the interaction between different organ systems, impact of injury and treatment. 1. Define conditions and materials to build the chip model.		

	2. Use multiphoton lithography bioprinting to develop microfluidics system that interconnects different organoids.		
Theme 2	Repair	Duration	2025-2027
Objective/ Activities	Comprehend molecular events that modulate tissue injury and repair in a multiple organ on chip system. 1. Explore effects of transplantation related injuries, such as hypoxia and inflammation, on different organoid systems. 2. Photoacoustic imaging for non-invasive assessment to determine organoid injury and repair.		
Theme 3	Regeneration	Duration	2026-2029
Objective/ Activities	Studying the effect of stem cell & organoid-based therapies on regeneration of non-functioning multi-organs on chip. 1. Exploring methodology to allow interaction of cellular therapies with multiorgans-on-a-chip.		
WP2	Development of perfusion platforms	Total Duration	2022-2030
Institute(s)	TU Delft/Erasmus MC/EUR		
Summary	We will develop technology to extend out-of-the-body time of donor organs (>7 days) and enable interventions (pharmacological and cell-based) to make unsuitable donor organs suitable for transplantation.		
Theme 1	Preservation	Duration	2022-2024
Objective/ Activities	Optimize long-term perfusion to maintain tissue viability and function. Refine functional assays to determine repair of ischemia induced damage.		
Theme 2	Repair	Duration	2024-2029
Objective/ Activities	Develop perfusion and sensor system platform enabling long-term <i>ex vivo</i> perfusion, extending time from 8 hours to 8 days.		
Theme 3	Regeneration	Duration	2025-2030
Objective/ Activities	Reconditioning damaged donor organs. 1. Using iPS generated autologous organ specific target cells. 2. Using in vivo organ specific gene editing systems. 3. Using autologous organoid/stem cell technology.		

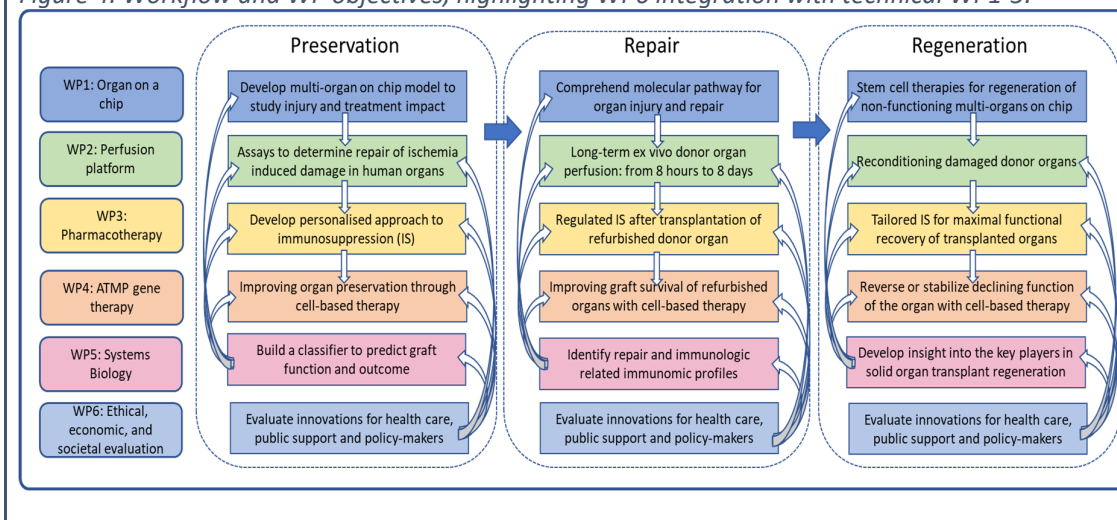
	4. Using growth factors and other agents (ATMP cell products) to stimulate regeneration.		
WP3	Pharmacotherapy to preserve and repair	Total Duration	2022–2029
Institute(s)	ErasmusMC/TU Delft/EUR		
Summary	Due to variable efficacy and toxicity of immunosuppressive (IS) therapy, a personalized IS therapy is needed. Prediction of rejection vs toxicity risk using home-monitored real-time drug levels and inflammatory biomarkers will be developed using computational algorithms to guide and adjust immunosuppression, thus significantly enhancing the quality of life of recipients and the longevity of the transplanted organ.		
Theme 1	Preservation	Duration	2022-2028
Objective/ Activities	Optimizing long term quality of transplanted organs by tailoring IS. 1. Develop sensor device combining measurements of rejection biomarkers and drug levels. 2. Computational algorithms allowing personalized drug dosages.		
Theme 2	Repair	Duration	2022-2029
Objective/ Activities	Regulating immunosuppression after transplantation of repaired donor organs 1. Measuring biomarkers indicating persistent donor organ cell damage 2. Measuring immune reactivity related to repair. 3. Defining adapted immunosuppressive drug schedules preventing rejection and toxicity. 4. Developing continuous dosing to optimize drug dosages. 5. Develop dosing algorithms and dynamic models to guide IS treatment.		
Theme 3	Regeneration	Duration	2022-2029
Objective/ Activities	Allowing maximal functional recovery of transplanted organs by selecting appropriate immunosuppressive therapy. Assessing adherence to home-monitoring and medication regime. 1. Clinical trials focusing on reduced-dosing of toxic immunosuppressants to ensure maximum functional recovery.		
WP4	ATMP (cell) therapy to repair and regenerate	Total Duration	2023–2031

Institute(s)	ErasmusMC/EUR		
Summary	Cell-based therapy is a promising tool for improving the outcome of solid organ transplantation. We will study cell-based therapy to prevent and diminish ischemia reperfusion injury (IRI) and allograft rejection, to reduce the use of immunosuppression and to induce repair and regeneration. Biomarker analysis (blood, urine, perfusion fluid, biopsy) and integrative spatial and single cell transcriptomics mapping of the allograft will be used to obtain mechanistic insight.		
Theme 1	Preservation	Duration	2023–2031
Objective/ Activities	Improving organ preservation through cell-based therapy 1. Testing cell-based therapy on a multiorgan-on-a-chip model (WP 1). 2. Administration of cell-based therapy to precondition perfused organs to prevent injury.		
Theme 2	Repair	Duration	2023–2031
Objective/ Activities	Improving long-term graft survival of repaired organs with cell-based therapy 1. Testing cell-based therapy on chip model to minimize immune suppression. 2. Administration of cell-based therapy after transplantation to minimize immunosuppression of refurbished donor organs. 3. demonstration of safety, feasibility, and efficacy of cell-based therapy of refurbished organ.		
Theme 3	Regeneration	Duration	2023–2031
Objective/ Activities	Reverse or stabilize declining function of the organ with cell-based therapy. 1. Testing cell-based therapy on a chip model to reverse fibrosis. 2. Administration of cell-based therapy to regenerate organs with damage. 3. Demonstration of safety, feasibility, and efficacy of cell therapy to induce repair of damaged organs.		
WP5	Monitoring of function, prognostication and clinical evaluation	Total Duration	2023–2032
Institute(s)	ErasmusMC/TU Delft		
Summary	Understanding tissue repair and regeneration to efficiently develop regenerative strategies. Detailed monitoring of transplant injury is vital to secure transplant outcome.		

Theme 1	Preservation	Duration	2023-2026
Objective/ Activities	Build a classifier based on noninvasive and tissue related injury and inflammatory markers to predict graft function and outcome (WP1). 1. Assessment of multi-omics in donor perfusion fluids and tissue biopsy material, and in patient blood. 2. Computational analysis to develop a functioning classifier.		
Theme 2	Repair	Duration	2026-2028
Objective/ Activities	To identify repair and immunologic related immunomic profiles discovered in WP5.1 1. Functional validation of immunomics profile and algorithm (WP5.1). 2. Validate classifier WP5.1 on for-cause transplant biopsies.		
Theme 3	Regeneration	Duration	2028-2032
Objective/ Activities	Develop insight into solid organ transplant regeneration. 1. Investigate tissue injury effects on (identified in WP5.1) immunological processes through lymph node-on-a-chip studies. 2. Investigate the effect of recipient derived inflammatory cells on donor derived endothelial cells (donor endothelial provocation test). 3. Real-time tissue imaging of lymph node for function cellular interaction. 4. Real-time tissue imaging of 3D vessel on a chip to visualize tissue regeneration.		
WP6	Evaluation of novel technologies and allocation options from ethical, legal, health economic, patient-centered, and societal perspectives.	Total Duration	2022-2032
Institute(s)	EUR/ErasmusMC/TUDeft		
Summary	The development of repaired donor organs requires answering of novel ethical and legal questions. WP6 will evaluate whether the Flagship approach agrees with the patient needs and represents value for money. WP6 is represented separately for clarity yet is an integral component of WP2-5.		
Theme 1	Preservation-Repair-Regeneration	Duration	2022-2032
Objective/ Activities	Evaluation of allocation of <i>ex-vivo</i> treated donor organs and the ethical, legal, health economic, patient, and societal consequences of advanced high-tech modular body part replacing healthcare. Activities include evaluation to determine:		

	1. The best approach for clinical implementation of repaired donor organs in an ethical way, necessary for good clinical decision-making for future patients and for policy making (e.g. allocation guidelines). 2. Ethical implications of the availability of 'repaired' organs for clinical care and for the distribution of organs (a scarce resource) in terms of allocation, indication and prioritization. 3. Guidance for the development of a) S(M)DM guideline (including decision aids) and b) systems of allocation that meets legal requirements concerning refurbished and other new options for donor organs. 4. Short- and long-term effects on physical and mental health outcomes, quality of life (QoL), and societal and healthcare costs for using repaired organs vs other alternatives. 5. Effects of extra donor organ supply on waiting lists, the inclusion of new patient groups, societal and individual health outcomes, QoL and costs, depending on allocation mechanisms for different organs. 6. Weighting of ethical, legal, clinical, economic, and patient preference aspects in an (early) Health Technology Assessment (HTA) concerning organ transplantation. 7. Tools to monitor the costs, (clinical) benefits and stakeholder preferences after organ transplantation.
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Figure 4: Workflow and WP objectives, highlighting WP6 integration with technical WP1-5.



3.d Alignment of the program with relevant national and international research agendas (71 words)

This Flagship is positioned at the center of the Nationale Wetenschapsagenda for health: How to enable vital aging, while dealing with a strong increase of chronic diseases. We address the challenge to develop universal therapies for different organ diseases.

Nationale Wetenschapsagenda specific themes: 95, 100, 101, 102 & 103

Health Holland LSH program roadmaps:

- No 2: imaging & image guided therapies
- No 4: regenerative medicine
- No 5: pharmacotherapy

2. Consortium (25%) max 2000/2000 words for 4.a to 4.d

4.a Track record of the consortium members (624 words)

3 Institutes and 9 Departments are involved, including researcher from EMC (7 clinicians, 9 researcher) EUR (7) and TU Delft (6). The consortium presents an impressive, combined track-record, demonstrating strong capabilities in clinical translation of scientific results, effective student engagement and raising funding. This section will focus on those with key roles within the project:

Prof. dr. Jan IJzermans (EMC; Flagship lead) - transplant surgeon, conducts preclinical and clinical research. Director of European educational transplant program, supervised >50 PhD's. director of the Erasmus MC Transplant Institute. Raised funding (> 5M; VWS) for structuring donor organ perfusion in Erasmus MC.

Prof. dr. Paddy French (TU Delft; Flagship co-lead), IEEE and IAAM-fellow, expert on integrated sensor systems, micromachining, for medical and environmental applications. Raised external funding for >50 projects, and established a strong network on medical devices, including catheters and premature baby treatment technology. Supervised 35 PhDs, >100 MSc, 10 BSc projects, is a lecturer in BSc and MSc since 1994 on sensors, actuators, displays and medical microsystems.

Prof. dr. Esther de Bekker (EUR; Flagship co-lead), Health Economics & Health Preferences, co-director of the interfaculty Erasmus Choice Modelling Centre. Expert in choice modelling, economic evaluation, cost, and benefit measurement, €6.1M in funding (NWO (Veni & Vidi laureate), IMI, Horizon2020, Erasmus Initiative, Research Excellence Initiative EUR, VGZ health insurance, Erasmus MC efficacy, EuroQoL, GSK, Sanofi, Convergence ESHPM, National Institute for Public Health). Best PhD supervisor award Erasmus Graduate School 2018, supervised 20 MSc students over the past five years.

Prof. dr. Luc van der Laan (EUR), expertise Liver Regenerative Medicine, Network Medical Delta, ILTS, ESOT, TERMIS, EASL, NTV; > 200 publications; h-factor=53; >1.M In funding. engaging & supervising BSc/ Msc students: Initiator of several Regenerative Medicine courses. Supervised >50 undergraduate and PhD students.

Dr. Massimo Mastrangeli (TU Delft), expert in micro/nanofabrication and design of organ-on-chip devices with integrated sensors and microactuators. His work Includes publications In prominent journals (Nature, Science, Nanotech, PNAS). He Is co-applicant In TTW Perspectief and ORC grants. Since joining TU Delft he supervise 11 MSc (5 ongoing) and 24 BSc thesis projects.

Dr. Jeroen de Jonge (EMC) transplant surgeon, realized the Organ Preservation Room in the Erasmus MC, a clean room harboring 4 organ perfusion / repair docks. Raised €1.6M in funding (TKI-LSH, VWS, ESOT, Erasmus MC). Spent 10 years as in Examination Board Medicine, 1 yr Querido Honours College and supervises BSc Clinical Technology/MDC/PhD students

Prof. Dirk-Jan G.M. Duncker (EMC) Experimental Cardiology, translational preclinical cardiovascular research, raised ~€7.1M in funding; (ZonMw, Dutch Heart Foundation, American Heart Association, Industry, Government, EU (FP7). involved in preclinical testing of devices and drugs in experimental models of cardiovascular disease and in valorization activities within DCVA-RECONNEXT consortium. Has supervised 20+ BSc/MSc students and 30+ PhD students.

Prof. dr. Reinders (EMC), nephrologist, expertise cell therapy; past President Dutch Transplant Society; Board member of Eurotransplant, The Transplant Society, and Scientific Board NSN, Chair ECTORS ESOT and Member ESOT Council; >120 scientific papers; ~€6 M in funding; PI of 3 clinical trials with mesenchymal cells. Set up of Leiden Oxford Transplantation Summerschool; teaching Ba/MA/ PhD's

Prof. dr. Wouter A. Serdijn (TU Delft) - experienced in microelectronics, biomedical electronics, bioelectronics, raised external funding: NWO-TTW Demonstrator, NWO Take-Off project, 3 large European ECSEL JU projects, Medical Delta programs, Medical NeuroDelta and Medical Delta Cardiac Arrhythmia Lab, Hartstichting-STW, 8 NWO-AES OTP projects. Prof. Serdijn supervised 39 BSc and 142 MSc students, is teaching at all levels of education and has an excellent track record in translation of results:

- 5 patents (of which 2 filed in the US, 1 in China, Japan, and Europe);
- 2 companies (NOWI and Momo Medical);

Prof. Hugo van der Kuy (EMC), pharmacology, research on efficacy and toxicity of drugs. ~€4M in funding; ZonMw/Senternovem/KNMP. Supervisors BSc, MSc, PhD students

4.b Expertise of the consortium members in relation to the aims and objectives of the program (364)

The program lead (Ijzermans) has a long lasting experience in organ transplantation and is embedded in translational and clinical transplant research. The EMC is the largest transplant center in the Netherlands and includes heart, lung, liver and kidney transplantation. Research focuses on organ specific and overarching aspects of transplantation, such as on immunology (Baan, Clahsen, Reinders), cellular repair and regeneration (Van der Laan, Gribnau, Kanaar) all having internationally acknowledged research programs. The EMC also has proven expertise regarding the ethical and societal aspects of organ donation. The EUR is leading in research in the field of policy, organizational sciences with a focus on health care, health economics, health technology assessment, health preference research, social medical sciences, health law and health insurance and will put the transplant innovations in societal perspective. (deBekker-Grob, Goossens, Donkers, Buijsen, Redekop) At TU Delft, technological knowhow is available on design and development of imaging techniques, (micro) sensors, state-of-art chemical analyzers and novel olfaction techniques to determine environmental changes (French, Serdijn, Mastrangeli, Ottens). So far, application of these techniques to the field of transplantation has not been accomplished.

Aligning the three knowledge institutes to address the problem of organ shortage on a generic level will unleash crucial research power, allowing to think outside the box of a specific organ. This convergence will create a platform that will aid clinicians through the efforts of applied and fundamental researchers and health legal, economic and ethics researchers. Relevant fundamental and technical issues as well as societal dilemmas, including organ shortage, waiting list mortality and waiting list access disparity for elderly people, being the most affected by organ failure, will be addressed. To reach the overall aim of the Organ Transplantation Flagship proposal (20% more organs transplanted through donor organ repair and regeneration), six specific key objectives were formulated, leading to six work packages that will integrate the expertise existing in the three knowledge institutes as appropriate for that work package. Specific milestones and deliverables for the work packages will be determined during joint meetings with all participants (co-leads and co-applicants) that will contribute to the aim of the Organ Transplantation Flagship. Additionally, many other collaborating experts to this Flagship are listed in section 4c.

4.c Involvement of partners (625 words)

Table 7: a list of partners including expertise and roles within WPs

WP	Consortium member	Expertise	Role in WP
1	L.van der Laan (EMC)	Liver regenerative medicine, Head of the Laboratory (LETIS) at EMC	WP-leader WP1
	M.Hoogduijn (EMC)	Medical biology: cell therapies, iPSC-derived organoids, organ-on-chip models.	scientific input, supervisor
	J.Essers (EMC)	Molecular imaging	scientific input, supervisor
	A.Zadpour (TUD)	3D/4D printing	scientific input, supervisor
	P.Boukany (TUD)	Microfluidics, biomechanics	scientific input, supervisor
	M.Mastrangeli (TUD)	micro/nanofabrication, organ-on-chip devices, microsensors	co-development of microfluidic devices
	J. Gribnau (EMC)	Stem cell research	Scientific input
	Y. Taverne	Translational Cardio Research	Scientific input
2	J.de Jonge (EMC)	Transplant Surgery, organ retrieval, organ perfusion	WP-leader WP2
	J.N.M. IJzermans (EMC)	Organ transplantation	Flagship leader and supervisor
	G.van Soest (EMC)	Minimally invasive imaging technology, light and ultrasound, translational research.	Supervisor
	R.Minnee (EMC)	Transplant Surgery, minimally invasive surgery, Clinical and experimental expertise in normothermic machine perfusion in human and animal kidneys	Scientific input
	E.Mahtab (EMC)	Cardiothoracic surgery	Scientific input
	O.Manintveld (EMC)	Transplant Cardiology	Scientific input
	R.Hoek (EMC)	Lung transplantation	Scientific input
	T.Goos (EMC)	Neonatology and incubator systems	Scientific input
	J.Essers (EMC)	Molecular imaging	Supervisor
	D.Duncker (EMC)	Cardiovascular (patho)physiology, experimental cardiology, preclinical translational research in experimental models of ischemic heart disease and heart failure.	Supervisor
	P.French (TUD)	Sensors, sensor technology, sensors in biomedical and environment applications	Project leader
	M.Mastrangeli (TUD)	As above (hereon marked "-")	Scientific input
	J.Dankelman (TUD)	Minimally invasive surgery, perfusions systems	Supervisor
	P.Boukanay (TUD)	Microfluidics, cell analysis on chip, biomechanics of tissues, biopolymers, micro-tissue, Organ-on-a-chip	Supervisor
	S.Iskander (TUD)	Photoacoustic imaging	Scientific input
	W.Serdijn (TUD)	Bioelectronics, Biomedical circuit and systems, Bioelectronic medicine	Scientific input
	E.deBekker-Grob (EUR)	Health Economics, Health Preferences, Economic Evaluation, Health Technology Assessment	EUR lead and supervisor
	M.Schermer (EMC)	Medical ethics, bioethics	Supervisor
	M.J. Mirzaali (TUD)	3D printing, biomimetics, bone mechanics, computational modeling	Supervisor

	M. Ottens (TUD)	High Throughput Process Development (HTPD) & Complex Molecules Separation; Chromatography	Scientific input
	H. van der Kuy	Pharmacist	WP leader WP 3 and supervisor
3	D.A.Hesselink (EMC)	Clinical nephrology, clinical pharmacology, transplant immunology	Scientific input
	M.E.J.Reinders (EMC)	Clinical nephrology, clinical transplantation, cell therapy, transplant immunology, translational medicine, psychosocial aspects and education	Scientific input
	C. Baan (EMC)	Transplantation Immunology	Scientific input
	L. van der Laan (EMC)	-	Scientific input
	M. Hoogduijn (EMC)	Medical biology: cell therapies, iPSC-derived organoids, and organ-on-chip models.	Scientific input, supervisor
	M. Mastrangeli (TUD)	micro/nanofabrication, organ-on-chip devices, microsensors	co-development of microfluidic devices
	P.French (TUD)	-	Project leader
	M.Schermer (EMC)	Medical ethics, bioethics	Supervisor
	M.A. Mankowski (EUR)	Transplantation access using computational modelling, Mathematical modelling of graft survival, Artificial Intelligence	Scientific input
	E.deBekker-Grob (EUR)	Health Economics, Health Preferences, Economic Evaluation, Health Technology Assessment	EUR lead and supervisor
	M.E.J.Reinders (EMC)	-	WP leader- WP4
4	D.A.Hesselink (EMC)	-	Scientific input
	M.Hoogduijn (EMC)	-	Scientific input, supervisor
	C. Baan (EMC)	-	Scientific input
	R. Kramann (EMC)	single cell RNA-sequencing, spatial multi-omics, drug target discovery & validation, bioinformatics, organ fibrosis, CKD	Scientific input
	L. van der Laan (EMC)	-	Scientific input
	J.de Jonge (EMC)	-	Scientific input
	E.deBekker-Grob (EUR)	-	EUR lead and supervisor
	M.Schermer (EMC)	-	Supervisor
	E.Massey (EMC)	Psychosocial aspects of transplantation, qualitative research on the patient perspective	Scientific input
	E. de Pater (EMC)	Transplant Hematology	Supervisor
	J. Gribnau (EMC)	Stem cell research	Scientific input
	J.N.M.IJzermans	Transplant Surgeon	WP Leader WP 5
5	C. Baan (EMC)	-	Scientific input
	M.C. Clahsen-van Groningen (EMC)	Renal Transplant Pathology and transcriptomics expertise	Supervisor
	A.Stubs (EMC)	Bioinformatics	Scientific input
	R. Hendriks (EMC)	Immunity of the lung	Supervisor
	J.de Jonge (EMC)	-	Scientific input
	R.Minnee (EMC)	-	Scientific input
	D.Duncker (EMC)	-	Supervisor

	M.A. Mankowski (EUR)	-	Scientific input
	R. Kanaar	Molecular genetics	Supervisor
6	E.deBekker-Grob (EUR)	-	WP 6 leader and supervisor
	B.Donkers (EUR)	Data science and consumer behaviour. Understanding human behaviour and underlying decision-making processes from an empirical perspective	Scientific input
	M.Schermer (EMC)	-	Scientific input
	L.Goossens (EUR)	Health technology assessment, epidemiology	Scientific input
	J.N.M. IJzermans (EMC)	-	Flagship Leader, supervisor
	K.Redekop (EUR)	Health technology assessment	Scientific input
	P.French (TUD)	-	Project leader
	M.E.J.Reinders (EMC)	-	Scientific input

4.d Governance, organization and collaboration strategy of the consortium (350 words)

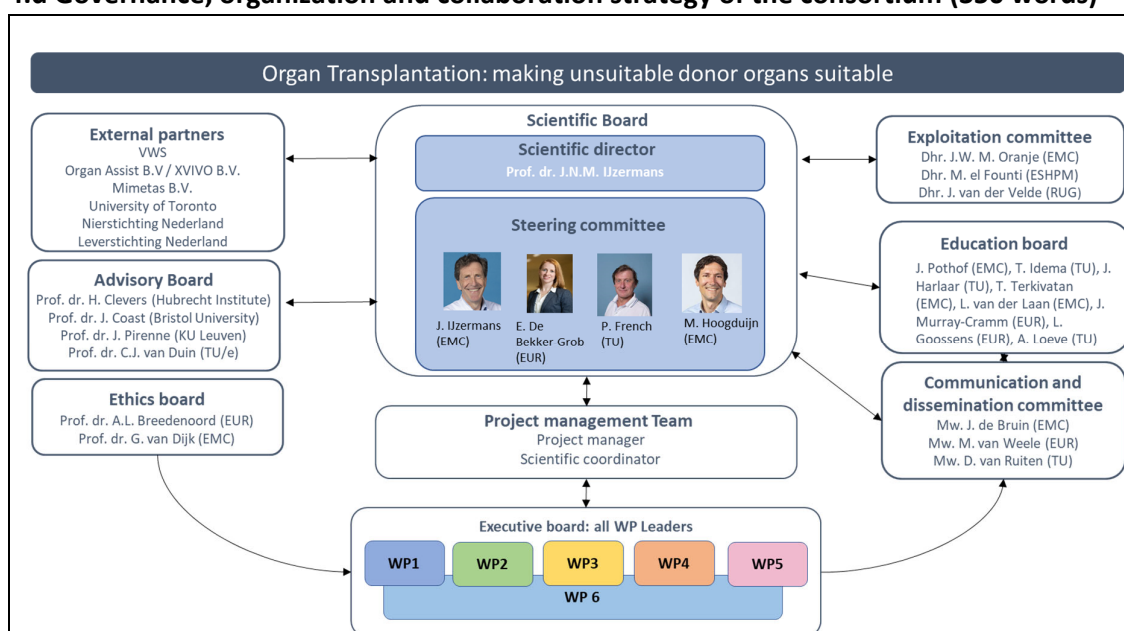


Figure 5: Project governance structure

The Organ Transplantation consortium is organized to facilitate collaboration and optimize expertise use of all consortium members. To mitigate the risk of losing cohesion, we plan a management structure as depicted in Figure 5. The bodies within the Organ Transplantation consortium are:

- **Scientific Board:** comprises the Scientific Director and Steering committee to oversee the overall strategy and deciding contingency plans.
- **Scientific Director:** will act as main lead of the Flagship (IJzermans).
- **Steering committee:** Main tasks include monitoring of WP progress, deliverables, budget, coordination of reporting.
- **Project management team:** three postdocs as scientific coordinators and a project manager responsible for the preparation and organization of consortium meetings and the advisory board, as well as maintaining efficient internal project documentation.
- **Advisory board:** provides independent scientific and ethics expert advice with meetings taking place once a year. The scientific advisory board includes:

- Hans van Duijn, full professor of Applied Analysis and former director of TU/e (rector magnificus).
- Hans Clevers, Prof. of molecular Genetics, UMCU and Hubrecht Institute, head research Roche Pharma.
- Jacques Pirenne, Prof. of Transplant Surgery University of Leuven, chair of the Belgian Transplant Society.
- Jo Coast, Prof. of Health Economics for Social Science and Medicine, University of Bristol, and Chair of the Scientific Committee for the International Health Economics Association.

The ethics advisory board includes:

- Annelien Breedenoord, Prof Ethics of biomedical science
- Gert van Dijk, ethicist, expertise on Organ Donation and Societal Impact
- **The WP Leaders** will be responsible for the day-to-day supervision, monitoring of task progress, coordination of collaboration, and timely achievement of deliverables within budgetary limits.
- **Communication and Dissemination team** will organize activities, disseminate findings to general audiences including social media outreach, and generate new routes towards impact for society
- **Education team** will organize half-yearly symposia for all team-members, colloquium series with external keynotes, and facilitate joint supervision for PhD and postdoc projects from at least two institutes.
- **Exploitation committee** will be responsible for coordinating partnerships and clinical translation of project results, especially through communication with **external partners** (industry).

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

A pluriannual plan will be required if your proposal is accepted

II Letters of intent and letters of commitment

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

III Letter of support from department head of the main lead

On headed paper of the department or institution.

IV References

Proposal references

1. Fiscella K, Williams DR. Health disparities based on socioeconomic inequities: implications for urban health care. Acad Med. 2004 Dec;79(12):1139–47.
2. Zhang Y, Gerdtham U-G, Rydell H, Jarl J. Socioeconomic Inequalities in the Kidney Transplantation Process: A Registry-Based Study in Sweden. Transplant Direct. 2018 Jan 24;4(2):e346.

3. Randhawa G. Achieving equality in organ donation and transplantation in the UK: challenges and solutions. :9.
4. A normothermic ex vivo organ perfusion delivery method for cardiac transplantation gene therapy | Scientific Reports [Internet]. [cited 2022 Feb 27]. Available from: <https://www.nature.com/articles/s41598-019-43737-y>
5. Stamp NL, Shah A, Vincent V, Wright B, Wood C, Pavey W, et al. Successful Heart Transplant after Ten Hours Out-of-body Time using the TransMedics Organ Care System. Heart, Lung and Circulation. 2015 Jun 1;24(6):611–3.
6. Yuzefovych Y, Valdivia E, Rong S, Hack F, Rother T, Schmitz J, et al. Genetic Engineering of the Kidney to Permanently Silence MHC Transcripts During ex vivo Organ Perfusion. Frontiers in Immunology [Internet]. 2020 [cited 2022 Feb 26];11. Available from: <https://www.frontiersin.org/article/10.3389/fimmu.2020.00265>

Key publications from consortium members

Roos FJM/Marsee A, Verstegen MMA; HPB Organoid Consortium, Gehart H, de Koning E, Lemaigre F, Forbes SJ, Peng WC, Huch M, Takebe T, Vallier L, Clevers H, Spee B/**van der Laan LJW**. Building consensus on definition and nomenclature of hepatic, pancreatic, and biliary organoids. Cell Stem Cell 2021, 28(5):816-832 (T10%, IF 24.6)

Roos FJM, Wu H, Willemse J, Lieshout R, Albarinos LAM, Kan YY, Poley JW, Bruno MJ, **de Jonge J**, Bártfai R, Marks H, **IJzermans JNM**, Verstegen MMA, **van der Laan LJW**. Cholangiocyte organoids from human bile retain a local phenotype and can repopulate bile ducts in vitro. Clin Transl Med. 2021 Dec;11(12):e566. doi: 10.1002/ctm2.566. PMID: 34954911

Roos FJM, Verstegen MMA, Muñoz Albarinos L, Roest HP, Poley JW, Tetteroo GWM, **IJzermans JNM**, **van der Laan LJW**. Human Bile Contains Cholangiocyte Organoid-Initiating Cells Which Expand as Functional Cholangiocytes in Non-canonical Wnt Stimulating Conditions. Front Cell Dev Biol. 2021 Feb 9;8:630492. doi: 10.3389/fcell.2020.630492. eCollection 2020. PMID: 33634107

Willemse J, Roos FJM, Voogt IJ, Schurink IJ, Bijvelds M, de Jonge HR, **van der Laan LJW**, **de Jonge J**, Verstegen MMA. Scaffolds obtained from decellularized human extrahepatic bile ducts support organoids to establish functional biliary tissue in a dish.

Biotechnol Bioeng. 2021 Feb;118(2):836-851. doi: 10.1002/bit.27613. Epub 2020 Nov 9. PMID: 33118611

Verstegen MMA, Roos FJM, Burka K, Gehart H, Jager M, de Wolf M, Bijvelds MJC, de Jonge HR, Ardisasmita AI, van Huizen NA, Roest HP, de Jonge J, Koch M, Pampaloni F, Fuchs SA, Schene IF, Luider TM, van der Doef HPJ, Bodewes FAJA, de Kleine RHJ, Spee B, Kremers GJ, Clevers H, **IJzermans JNM**, Cuppen E, **van der Laan LJW**. Human extrahepatic and intrahepatic cholangiocyte organoids show region-specific differentiation potential and model cystic fibrosis-related bile duct disease.

Sci Rep. 2020 Dec 14;10(1):21900. doi: 10.1038/s41598-020-79082-8. PMID: 33318612

M. Mastrangeli; S. Millet; the ORCHID partners; J. van den Eijnden-van Raaij, "Organ-on-Chip in Development: Towards a roadmap for organs-on-chip", ALTEX-Alternatives to Animal Experimentation, Volume 36, Issue 4, pp. 650-668, 2019. DOI: 10.14573/altex.1908271

M. Mastrangeli; S. Millet; C. Mummery; P. Loskill; D. Braeken; W. Eberle; M. Cipriano; L. Fernandez; M. Graef; X. Gidrol; N. Picollet-D'Hahan; B. van Meer; I. Ochoa; M. Schutte; J. van den Eijnden-van Raaij, " Building Blocks for a European Organ-on-Chip Roadmap", ALTEX - Alternatives to Animal Experimentation, Volume 36, Issue 3, pp. 481-492, 2019. DOI: 10.14573/altex.1905221

M. Dostanic; L. M. Windt; J. M. Stein; B. J. van Meer; M. Bellin; V. Orlova; **M. Mastrangeli**; C. L. Mummery; P. M. Sarro, " A miniaturized EHT platform for accurate measurements of tissue contractile properties", Journal of Microelectromechanical Systems, Volume 29, Issue 5, pp. 881-887, July 2020. DOI:10.1109/JMEMS.2020.3011196

H. Aydogmus; H. J. van Ginkel; A.-D. Galiti; M. Hu; J.-P. Frimat; A. van den Maagdenberg; G.Q. Zhang; **M. Mastrangeli**; P. M. Sarro, " Dual-gate FET-based charge sensor enhanced by in-situ electrode decoration in a MEMS organs-on-chip platform", In 21st Int. Conf. on Solid-State Sensors, Actuators and Microsystems (TRANSDUCERS 2021), online, 20-25 June, 2021.

Nele Revyn; Michel. H. Y. Hu; Jean-Philippe M. S. Frimat; B. de Wagenaar; Arn M. J. M. van den Maagdenberg; Pasqualina M. Sarro; **Massimo Mastrangeli**, " Recording neuronal activity on chip with segmented 3D microelectrode arrays", In 35th Intl. Conf. on Micro Electro Mechanical Systems (MEMS 2022), January 9-13 2022.

Wenqi Hu , Guo Zhan Lum , **Massimo Mastrangeli** , Metin Sitti . Small-scale soft-bodied robot with multimodal locomotion. Nature. 2018 Feb 1;554(7690):81-85. doi: 10.1038/nature25443. Epub 2018 Jan 24.

Akgun OC, Nanbakhsh K, Giagka V, **Serdijn WA**.
A Chip Integrity Monitor for Evaluating Moisture/Ion Ingress in mm-Sized Single-Chip Implants.
IEEE Trans Biomed Circuits Syst. 2020 Aug;14(4):658-670. doi: 10.1109/TBCAS.2020.3007484. Epub 2020 Jul 7. PMID: 32746351

Rezaeiyan Y, Zamani M, Shoaie O, **Serdijn WA**.
Mixed-Signal IC With Pulse Width Modulation Wireless Telemetry for Implantable Cardiac Pacemakers in 0.18- μ m CMOS.
IEEE Trans Biomed Circuits Syst. 2018 Jun;12(3):589-600. doi: 10.1109/TBCAS.2018.2819021. PMID: 29877822

Steijlen ASM, Bastemeijer J, Groen WA, Jansen KMB, **French PJ**, Bossche A. Development of a microfluidic collection system to measure electrolyte variations in sweat during exercise. Annu Int Conf IEEE Eng Med Biol Soc. 2020 Jul;2020:4085-4088. doi: 10.1109/EMBC44109.2020.9176123. PMID: 33018896

Roos FJM, Bijvelds MJC, Verstegen MMA, Roest HP, Metselaar HJ, Polak WG, Jonge HR, **IJzermans JNM, van der Laan LJW**. Impact of hypoxia and AMPK on CFTR-mediated bicarbonate secretion in human cholangiocyte organoids. Am J Physiol Gastrointest Liver Physiol. 2021 May 1;320(5):G741-G752. doi:10.1152/ajpgi.00389.2020. Epub 2021 Mar 3. PMID: 33655768

Schlegel A, van Reeve M, Croome K, Parente A, Dolcet A, Widmer J, Meurisse N, De Carlis R, Hessheimer A, Jochmans I, Mueller M, van Leeuwen OB, Nair A, Tomiyama K, Sherif A, Elsharif M, Kron P, van der Helm D, Borja-Cacho D, Bohorquez H, Germanova D, Dondossola D, Olivieri T, Camagni S,

Gorgen A, Patrono D, Cescon M, Croome S, Panconesi R, Carvalho MF, Ravaioli M, Caicedo JC, Loss G, Lucidi V, Sapisochin G, Romagnoli R, Jassem W, Colledan M, De Carlis L, Rossi G, Di Benedetto F, Miller CM, van Hoek B, Attia M, Lodge P, Hernandez-Alejandro R, Detry O, Quintini C, Oniscu GC, Fondevila C, Malagó M, Pirenne J, **Ijzermans JNM**, Porte RJ, Dutkowski P, Taner CB, Heaton N, Clavien PA, Polak WG, Muiesan P; DCD Collaborator Group.

A multicentre outcome analysis to define global benchmarks for donation after circulatory death liver transplantation. *J Hepatol*. 2022 Feb;76(2):371-382. doi: 10.1016/j.jhep.2021.10.004. Epub 2021 Oct 14. PMID: 34655663

Matton APM, Selten JW, Roest HP, **de Jonge J, Ijzermans JNM**, de Meijer VE, Porte RJ, **van der Laan LJW**. Cell-free microRNAs as early predictors of graft viability during ex vivo normothermic machine perfusion of human donor livers.

Clin Transplant. 2020 Mar;34(3):e13790. doi: 10.1111/ctr.13790. Epub 2020 Feb 20. PMID: 31984571

van Rijn R, Schurink IJ, de Vries Y, van den Berg AP, Cortes Cerisuelo M, Darwish Murad S, Erdmann JI, Gilbo N, de Haas RJ, Heaton N, van Hoek B, Huurman VAL, Jochmans I, van Leeuwen OB, de Meijer VE, Monbaliu D, Polak WG, Slangen JJG, Troisi RI, Vanlander A, **de Jonge J**, Porte RJ; Hypothermic Machine Perfusion in Liver Transplantation - A Randomized Trial DHOPE-DCD Trial Investigators. *N Engl J Med*. 2021 Apr 15;384(15):1391-1401. (IF 91.2) consortium **J Ijzermans**

Jochmans I, Brat A, Davies L, Hofker HS, van de Leemkolk FEM, Leuvenink HGD, Knight SR, Pirenne J, Ploeg RJ; Oxygenated versus standard cold perfusion preservation in kidney transplantation (COMPARE): a randomised, double-blind, paired, phase 3 trial COMPARE Trial Collaboration and Consortium for Organ Preservation in Europe (COPE). *Lancet*. 2020 Nov 21;396(10263):1653-1662. (IF 79.3) COPE consortium **J.de Jonge**

Schurink IJ, de Haan JE, Willemse J, Mueller M, Doukas M, Roest H, de Goeij FHC, Polak WG, **Ijzermans JNM**, Dutkowski P, **van der Laan LJW, de Jonge J**. A proof of concept study on real-time LiMax CYP1A2 liver function assessment of donor grafts during normothermic machine perfusion.

Sci Rep. 2021 Dec 6;11(1):23444. doi: 10.1038/s41598-021-02641-0. PMID: 34873187

Verstegen MMA, Willemse J, van den Hoek S, Kremers GJ, Luider TM, van Huizen NA, Willemssen FEJA, Metselaar HJ, **Ijzermans JNM, van der Laan LJW, de Jonge J**. Decellularization of Whole Human Liver Grafts Using Controlled Perfusion for Transplantable Organ Bioscaffolds.

Stem Cells Dev. 2017 Sep 15;26(18):1304-1315. doi: 10.1089/scd.2017.0095. Epub 2017 Jul 31. PMID: 28665233

Reinders MEJ, Groeneweg KE, Hendriks SH, Bank JR, Dreyer GJ, de Vries APJ, van Pel M, Roelofs H, Huurman VAL, Meij P, Moes DJAR, Fibbe WE, Claas FHJ, Roelen DL, van Kooten C, Kers J, Heidt S, Rabelink TJ, de Fijter JW. Autologous bone marrow-derived mesenchymal stromal cell therapy with early tacrolimus withdrawal: The randomized prospective, single-center, open-label TRITON study *Am J Transplant*. 2021 Sep;21(9):3055-3065. doi: 10.1111/ajt.16528. Epub 2021 Mar 18 (IF 8)

de Kok MJ, Schaapherder AF, Mensink JW, de Vries AP, **Reinders ME**, Konijn C, Bemelman FJ, van de Wetering J, van Zuilen AD, Christiaans MH, Baas MC, Nurmohamed AS, Berger SP, Ploeg RJ, Alwayn IP, Lindeman JH. A nationwide evaluation of deceased donor kidney transplantation indicates detrimental consequences of early graft loss. *Kidney Int*. 2020 Jun;97(6):1243-1252. doi: 10.1016/j.kint.2020.01.043. IF 10.6)

Hullegie-Peelen DM, Van Der Zwan M, Clahsen-Van Groningen MC, Mustafa DAM, Baart SJ, **Reinders MEJ**, Baan CC, Hesselink DA. Clinical and molecular profiling to develop a potential prediction model for the response to alemtuzumab therapy for acute kidney transplant rejection. Clin Pharmacol Ther 2022 Feb 24. doi: 10.1002/cpt.2566. Online ahead of print. PMID: 35202481 (IF 1.9)

Hoogduijn MJ, Issa F, Casiraghi F, **Reinders MEJ**. Cellular therapies in organ transplantation. Transpl Int. 2021 Feb;34(2):233-244. doi: 10.1111/tri.13789. Epub 2021 Jan 15. PMID: 33207013

Hoogduijn MJ, Montserrat N, van der Laan LJW, Dazzi F, Perico N, Kastrup J, Gilbo N, Ploeg RJ, Roobrouck V, Casiraghi F, Johnson CL, Franquesa M, Dahlke MH, Massey E, Hosgood S, **Reinders MEJ**. The emergence of regenerative medicine in organ transplantation: 1st European Cell Therapy and Organ Regeneration Section meeting. Transpl Int. 2020 Aug;33(8):833-840. doi: 10.1111/tri.13608. Epub 2020 Apr 28. PMID: 32237237

de Jong PGM, Hendriks RA, Luk F, Dos Santos ACS Jr, **Reinders MEJ**. Development and application of a massive open online course to deliver innovative transplant education. Transpl Immunol. 2021 Jun;66:101339. doi: 10.1016/j.trim.2020.101339. Epub 2020 Oct 24. PMID: 33164772

Meucci MC, **Reinders MEJ**, Groeneweg KE, Bezstarosti S, Ajmone Marsan N, Bax JJ, De Fijter JW, Delgado V. Cardiovascular Effects of Autologous Bone Marrow-Derived Mesenchymal Stromal Cell Therapy With Early Tacrolimus Withdrawal in Renal Transplant Recipients: An Analysis of the Randomized TRITON Study. J Am Heart Assoc. 2021 Dec 21;10(24):e023300. doi: 10.1161/JAHA.121.023300. Epub 2021 Dec 16. PMID: 34913362

Tejeda-Mora H, Leon LG, Demmers J, Baan CC, **Reinders MEJ**, Bleck B, Lombardo E, Merino A, Hoogduijn MJ. Proteomic Analysis of Mesenchymal Stromal Cell-Derived Extracellular Vesicles and Reconstructed Membrane Particles. Int J Mol Sci. 2021 Nov 29;22(23):12935. doi: 10.3390/ijms222312935. PMID: 34884740

van Reeve M, Gilbo N, Monbaliu D, van Leeuwen OB, Porte RJ, Ysebaert D, van Hoek B, Alwayn IPJ, Meurisse N, Detry O, Coubeau L, Cicarelli O, Berrevoet F, Vanlander A, **Ijzermans JNM**, Polak WG. Evaluation of Liver Graft Donation After Euthanasia JAMA Surg. 2020 Oct 1;155(10):917-924. doi: 10.1001/jamasurg.2020.2479. PMID: 32777007

van der Linden R, **Schermer M**. Health and disease as practical concepts: exploring function in context-specific definitions. Med Health Care Philos. 2022 Mar;25(1):131-140. doi: 10.1007/s11019-021-10058-9. Epub 2021 Nov 16. PMID: 3478397

Ehni HJ, Kadi S, **Schermer M**, Venkatapuram S. Toward a global geroethics - gerontology and the theory of the good human life. Bioethics. 2018 May;32(4):261-268. doi: 10.1111/bioe.12445. PMID: 29676503

Vos IML, **Schermer MHN**, Bolt ILLE. Recent insights into decision-making and their implications for informed consent.

J Med Ethics. 2018 Nov;44(11):734-738. doi: 10.1136/medethics-2018-104884. Epub 2018 Jul 21.

PMID: 30032106

Verweij M, van de Vathorst S, **Schermer M**, Willems D, de Vries M. Ethical Advice for an Intensive Care Triage Protocol in the COVID-19 Pandemic: Lessons Learned from The Netherlands.

Public Health Ethics. 2020 Sep 3;13(2):157-165. doi: 10.1093/phe/phaa027. eCollection 2020 Jul.

PMID: 33288984

Bekker - Grob, E.W. de, Donkers, B., Bliemer, M., Veldwijk, J. & Swait, J. (2020). Can healthcare choice be predicted using stated preference data? Social Science & Medicine, 246, 112736. (IF 4.2)

Bekker-Grob, E.W. de, Swait, J., Kassahun, H.T., Bliemer, M., Jonker, M.F., Veldwijk, J., Cong, K., Rose, J.M. & Donkers, B. (2019). Are Healthcare Choices Predictable? The Impact of Discrete Choice Experiment Designs and Models. Value in Health, 22 (9), 1050-1062. doi: 10.1016/j.jval.2019.04.1924 (IF 5.3)

Husereau D, Drummond M, Augustovski F, **de Bekker-Grob EW**. et al. Consolidated Health Economic Evaluation Reporting Standards 2022 (CHEERS 2022) statement: updated reporting guidance for health economic evaluations. Value in Health. 2022 p.10-31. (IF 5.3)

Veldwijk, J., Viberg Johansson, J., Donkers, B. & **Bekker - Grob, E.W. de** (2020). Mimicking Real-Life Decision Making in Health: Allowing Respondents Time to Think in a Discrete Choice Experiment. Value in Health, 23 (7), 945-952. doi: <https://doi.org/10.1016/j.jval.2020.02.014> (IF 5.3)

Huls, S.P.I., Whichello, C.L., Exel, N.J.A. van, Uyl-de Groot, C.A. de & **Bekker-Grob, E.W. de** (2019). What Is Next for Patient Preferences in Health Technology Assessment? A Systematic Review. Value in Health, 22 (11), 1318-1328. doi: <https://doi.org/10.1016/j.jval.2019.04.1930> (IF. 5.3)

Soekhai, V.R., **Bekker - Grob, E.W. de**, Ellis, A.R. & Vass, C.M. (2019). Discrete Choice Experiments in Health Economics: Past, Present and Future. Pharmacoeconomics (Print), 37 (2), 201-226. doi: <https://doi.org/10.1007/s40273-018-0734-2>

Whichello, C., Overbeeke, E. van, Janssens, R., Schölin Bywall, K., Russo, S., Veldwijk, J., Cleemput, I., Juhaeri, J., Levitan, B., Kübler, J., Smith, M., Hermann, R., Engelbrecht, M., Hueber, A., Comaneuscu, A., Harding, S., Simoens, S., Huys, I. & **Bekker - Grob, E.W. de** (2019). Factors and situations affecting the value of patient preference studies: semi-structured interviews in Europe and the US. Frontiers in pharmacology. doi: <https://doi.org/10.3389/fphar.2019.01009>

Jonker, M.F., Donkers, B., **Bekker - Grob, E.W. de** & Stolk, E.A. (2018). Advocating a Paradigm Shift in Health-State Valuations: The Estimation of Time-Preference Corrected QALY Tariffs. Value in Health, 21 (8), 993-1001. doi: <https://doi.org/10.1016/j.jval.2018.01.016>

Bekker-Grob, E.W. de & Chorus, C.G. (2013). Random Regret-Based Discrete-Choice Modelling: An Application to Healthcare. Pharmacoeconomics (Print), 31 (7), 623-634. doi: <https://doi.org/10.1007/s40273-013-0059-0>

Prof.dr. Jan N.M. IJzermans
Main Lead Flagship Organ Transplantation
Erasmus MC
Rotterdam

Groningen, March 14, 2022
Subject: Letter of Support to the
Convergence initiative
Ref: ub 2422

Dear prof. IJzermans,

With this letter I declare on behalf of XVIVO b.v. our interest in and support for the Convergence initiative, Flagship proposal 'Organ Transplantation'. We hereby approve that the XVIVO contribution, *in cash* and *in kind*, in the joined TKI project 'The Matrix Reloaded' may be used as matching for your proposal. This amount is defined as € 187.675 total for the project.

XVIVO focuses on the design, development, marketing and servicing of organ machine perfusion devices and disposable tubing sets. XVIVO has developed a broad portfolio of ex vivo perfusion devices for liver, kidney, heart, lung and donor perfusion.

XVIVO has a mission to stay on the forefront of making innovative products and developments available for their patients. The Convergence-project with the flagship proposal 'Organ Transplantation' perfectly fits in this mission. Based on the results of the project, organ perfusion is expected to be significantly improved leading to better outcomes. Patients receiving a donated organ will benefit from this in better and faster recovery, less complications and shorter hospital stay.

Yours sincerely,

Dr. Arjan van der Plaats
Director Research & Development

Prof. Dr. Marlies E.J. Reinders MD, PhD
Internist-nefroloog
Erasmus MC Transplantatie Instituut management team
Postbus 2040
3000 CA Rotterdam

Bussum, February 28, 2022

Referentie: WK/22.011

Subject: Letter of Support to project 'Organ Transplantation-making unsuitable donor organs suitable for transplantation'

Dear Prof. Dr. Reinders,

With this letter I declare on behalf of the Dutch Kidney Patient Association (NVN), our interest in and support for your research project 'Organ Transplantation-making unsuitable donor organs suitable for transplantation' that will be submitted in the Convergence-Health & Technology Flagship call.

The NVN is an active patient organisation established for and by patients with kidney disease, their relatives and kidney donors. We have more than 7,000 members and over 170 day-to-day volunteers: we provide a strong representation of kidney patients in the Netherlands. Our mission is to allow people with CKD to reach a higher quality of living, receive excellent care, and participate in society as well as support them in self-management. Our mission also includes ensuring that new research, policy and care actually benefits the patients for whom it is developed.

The NVN recognizes the importance of your proposal. Long waiting list for donor organs lead to the risks for patients and therefore reduced quality of life of many patients. This project aims to develop opportunities for better recovery and regeneration of suboptimal organs and is a promising step in reducing waiting list and thereby increased quality of life of many patients.

Yours sincerely,



Wanda Konijn
Konijn@nvn.nl
Teamcoördinator Quality & Research
Dutch Kidney Patient Association (NVN)



Hubrecht Institute

Developmental Biology
and Stem Cell Research

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Prof. Jan N.M. IJzermans MD, PhD
Head HPB & Transplant Surgery
Chair Erasmus MC Transplant Institute
P.O. Box 2040
3000 CA Rotterdam

Datum : 11 February 2022
Ref # : 20220114/HC/ave

Geachte heer IJzerman, beste Jan,

Naar aanleiding van je email van 10 februari jongstleden, bevestig ik mijn deelname aan het toekomstig samenwerkingsverband voor "Health and Technology".

Zoals aangegeven, zal ik met name online deelnemen in verband met mijn nieuwe positie bij Roche in Basel met ingang van 18 maart 2022.

Ik kijk uit naar de eerste bijeenkomst met alle experts.

Met vriendelijke groet,

Hans C. Clevers, MD, PhD
Professor Molecular Genetics at University Utrecht
Principal Investigator at Hubrecht Institute of the Royal Netherlands Academy
and at Prinses Máxima Centrum for Pediatric Oncology

Jos Joore
CEO MIMETAS BV
De Limes 7
2342 DH Oegstgeest

SUBJECT: Letter of Intent for the project "Organ Transplantation - Making unsuitable donor organs suitable for transplantation"

Utrecht, March 5, 2022

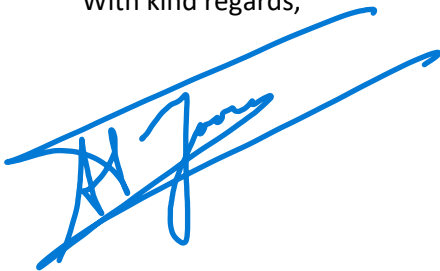
Dear Prof IJzermans,

I, Jos Joore, in my capacity of CEO at MIMETAS BV, hereby confirm that MIMETAS BV intends to contribute to the "Organ Transplantation - Making unsuitable donor organs suitable for transplantation" Flagships of Convergence project, as applied for by the main applicant, Jan IJzermans, Professor at Erasmus Medical Center, Erasmus University and Technical University Delft.

The applicants have taken the initiative to set up a partnership for "Health and technology" (<https://convergence.nl/healthtechnology>). This initiative brings together engineers, biomedical scientists and clinicians to develop new biotechnological plans for the benefit of patient care. The aforementioned project proposal is part of this initiative.

If the proposal gets awarded and the program will be conducted, MIMETAS intends to contribute in kind towards the project costs in accordance with the budget in the project proposal and budget form.

With kind regards,



Jos Joore
CEO MIMETAS BV
j.joore@mimetas.com
+31 6 516 96 884

Erasmus MC
t.a.v. prof. dr. J.N.M. IJzermans
Voorzitter Erasmus Transplantatie Instituut
Erasmus MC
Postbus 2040
3000 CA Rotterdam

Hoogland 18 februari 2022
Ref: 2022.02.34/JW

Geachte professor IJzermans,

Met grote belangstelling heeft de Nederlandse Leverpatiënten Vereniging (NLV) kennis genomen van het voornemen van het Erasmus MC Transplantatie Instituut een nauw samenwerkingsverband aan te gaan met de Technische Universiteit Delft en de Erasmus Universiteit, gericht op *Health and Technology*. De plannen voor een convergence wetenschapsgemeenschap (Flagship), samen met genoemde instituten verheugen ons zeer. Wij zijn ervan overtuigd dat dit alle processen rondom de zorg voor en van de leverpatiënt zal bespoedigen.

De NLV is al bij vele ontwikkelingen betrokken die betrekking hebben op diagnostiek, behandelinterventies, evenals de ontwikkelingen bij levertransplantatie waar veel leverpatiënten uiteindelijk mee in aanraking zullen komen. Wij weten dat er al goede ontwikkelingen gaande zijn zoals bredere inzetbaarheid van de perfusiesystemen ten einde levers voor transplantatie te optimaliseren, maar ook ontwikkeling van biosensoren, afbeeldende technieken, feedback modaliteiten en de toepassing van organoïden en iPSC. Een samenwerkingsverband zullen onder meer deze en nieuwe ontwikkelingen alleen maar ten goede komen.

Alleen door samenwerking met alle facetten van de wetenschap kan het uiteindelijke doel worden bereikt: het verbeteren van de kwaliteit van leven voor patiënten. Kortom, wij steunen dit fantastische initiatief van harte.

Met vriendelijke groet,
mede namens het bestuur



drs. José A. Willemse
directeur

Prof. Dr Marlies E.J. Reinders MD, PhD
Internist-nefroloog
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NL70INGB0000088000

Date February 17, 2022
Tel (+31) 35 697 80 15
E-mail research@nierstichting.nl
Concerns Flagship of convergence project proposal 'Organ Transplantation-making unsuitable donor organs suitable for transplantation'

LETTER OF SUPPORT

On behalf of the Dutch Kidney Foundation I would like to express our support for the project proposal 'Organ Transplantation-making unsuitable donor organs suitable for transplantation' that will be submitted in the Convergence Health & Technology Flagship call.

Our mission is to increase the quality of life of kidney patients and ultimately to cure kidney disease. Organ transplantation improves patient survival and quality of life and has a major beneficial impact on public health and the socio-economic burden of organ failure. Nevertheless, an unacceptably large number of people is waiting for a kidney and average waiting time is still too long because of the lack of suitable donor organs. Due to the lack of donor organs either people die on the waiting list or their condition deteriorated as such that transplantation is no longer possible.

This Flagship project holds great promise as the development of biosensors, imaging techniques, regulation of immunosuppressive medication, feedback modalities and the application of organoids and iPSC enables better recovery and regeneration of suboptimal donor kidneys through 'organ-on-a-chip models and machine perfusion systems.

Yours Sincerely,



Tom Oostrom
Director



Ministerie van Volksgezondheid,
Welzijn en Sport

> Retouradres Postbus 20350 2500 EJ Den Haag

Erasmus MC
T.a.v. de heer IJzermans
Postbus 2040
3000 CA Rotterdam

E-mail: c.zandijk@erasmusmc.nl

Datum 24 februari 2022
Betreft Letter of Support Flagship Organ Transplantation

Geachte heer IJzermans,

Met veel belangstelling heb ik uw brief gelezen over uw ambities voor een Flagship 'Organ Transplantation – making unsuitable organs suitable'. U geeft aan dat met de oprichting van een Flagship, het Erasmus MC, Erasmus universiteit, en de TU Delft hun expertises zullen bundelen om tot technologische ontwikkelingen te komen voor betere patiëntenzorg.

Machine-preservatie van organen die anders niet gedoneerd kunnen worden, met als doel die alsnog geschikt te maken voor transplantatie, is een belangrijke ontwikkeling om meer patiënten op de wachtlijst te kunnen helpen. VWS heeft daarom ook de verschillende initiatieven gesteund die u in uw brief noemt, en doet dit nog steeds. Nieuwe technologische ontwikkelingen en aanpassingen in machine-preservatie kunnen de kwaliteit van marginale organen verder verbeteren, waardoor meer organen gedoneerd en getransplanteerd kunnen worden. Ik onderschrijf dan ook het belang van uw ambities voor de Flagship 'Organ Transplantation – making unsuitable organs suitable'.

Hoogachtend,

de Minister van Volksgezondheid,
Welzijn en Sport,
namens deze,
de directeur Geneesmiddelen en Medische Technologie,

mw.mr. K.H.M. van Rooijen

Directoraat-Generaal

Curatieve Zorg

directie Geneesmiddelen en
Medische Technologie

Bezoekadres

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Inlichtingen bij

Guo, X. (Lili)
Senior beleidsmedewerker

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Kenmerk

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Bijlage(n)

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*Correspondentie uitsluitend
richten aan het retouradres
met vermelding van de
datum en het kenmerk van
deze brief.*

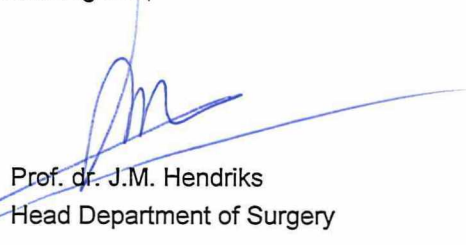
TO WHOM IT MAY CONCERN

Direct dial +31 (0)10 - 7031810
Fax number +31 (0)10 - 7032396
Internal postal address RG-219
E-mail c.zandijk@erasmusmc.nl
Our reference CE
Date March, 14th 2022

Dear Sir/ Madam,

As Head of the Department of Surgery I fully support the Scientific Director of the Flagship
"Organ Transplantation; making unsuitable donor organs suitable".

Kind regards,



Prof. dr. J.M. Hendriks
Head Department of Surgery

Postal address
P.O. box 2040
3000 CA Rotterdam, NL

Visiting address
Dr. Molewaterplein 40
3015 GD Rotterdam, NL

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Prof.dr. J.N.M. IJzermans
Head HPB and Transplant Program

Staff members
Prof.dr. C.H. J. van Eijck
Drs. M.U. Boehnert
Dr. B. Groot Koerkamp
Dr. J. de Jonge
Drs. H.J.A.N. Kimenai
Dr. R.C. Minnee
Dr. W.G. Polak
Dr. T. Terkivatan
Drs. T.C.K. Tran
Dr. R. F. de Wilde

Fellows
Dr. M. Cappelle
Drs. R. Haen
Dr. J. Shapiro

Date

March 22, 2022

Subject

Letter of intent

LETTER OF INTENT

for the

Organ Transplantation: Making Unsuitable Donor Organs Suitable

Convergence Flagship

Our reference

SW/20222203edb

Page

1/1

Visiting address

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E debekker-grob@eshpm.eur.nl

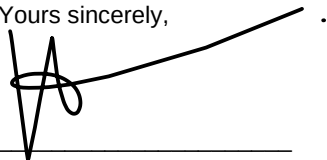
W www.eur.nl/eshpm

Dear professor Jan IJzermans (Erasmus MC),

I, prof. Werner Brouwer, in my capacity of pro Dean at Erasmus School of Health Policy & Management (ESHPM) hereby confirm that ESHPM intends to contribute to the '*Organ Transplantation: Making Unsuitable Donor Organs Suitable*' Convergence Flagship, as applied for by the main Flagship lead, professor J.N.M. IJzermans.

ESHPM will contribute to this flagship program with the awarded NWO Vidi project entitled 'Towards Accurate Prediction of Healthcare Choices' (2021-2026; led by the Flagship co-lead prof.dr. E.W. de Bekker-Grob), and with the awarded ESHPM PhD-project entitled 'Diabolical Dilemmas in Allocating Scarce Health Resources: Towards a Moral Choice Paradigm' (2020-2024; a joint project between ESHPM-TUD-EMC).

Yours sincerely,



Name: prof. W. Brouwer

Position: pro Dean

Date: 22 March 2022

Budget formulier: Convergentie Flagship Call

Flagship:	Eigen Transplantatie: making sustainable donor organs available
Flagship acronym:	
Naam Persoon:	2/3/2018, Grootman
Startdatum en looptijd van het project:	start: 1 oktober 2020, looptijd 3 jaar
Totaal project budget (incl. related projects):	€ 4.071.000
Eigen bijdragen en financiering:	€ 2.071.000
Aangevraagde Flagship financiering	€ 2.000.000

R&D Overview: kostensoorten per type onderzoek (hele project periode)

[illegible][illegible]

[illegible]

type	width	decimals	scaling	format
float	6	3275	000000	0000000000
percentage	000000	000000	000000	000000
currency	000000	000000	000000	000000
type	width	decimals	scaling	format
integer	000000			
decimal	000000			

The expanded convergence budget includes hiring of 6.1 FTEs, 3.7% of which are PhD students and 1.0 Fte Post Doc and in addition a quarter master will be appointed.